

22/01/09

DETERMINATION OF Z PARAMETERS OF TWO PORT NETWORK

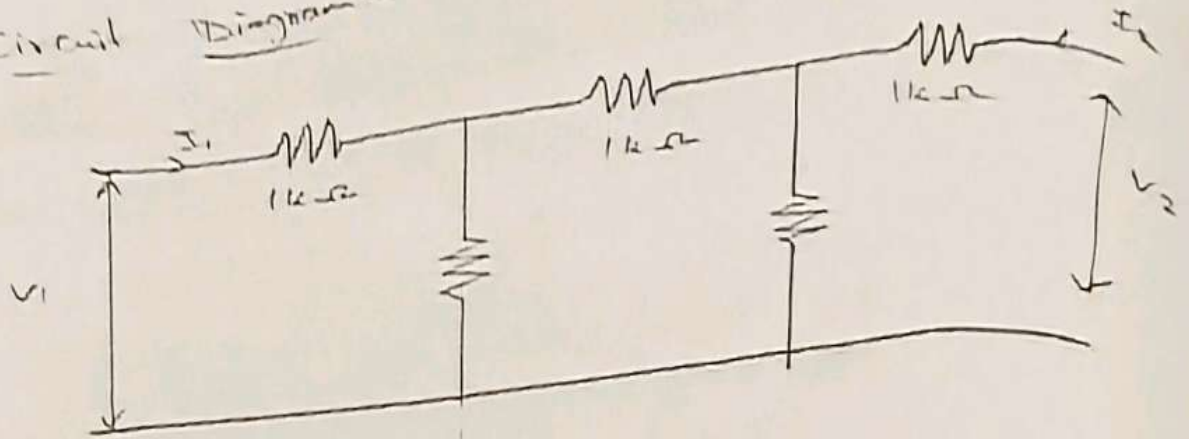
AIM:

To calculate and verify Z parameters of a two port network using multisim simulator.

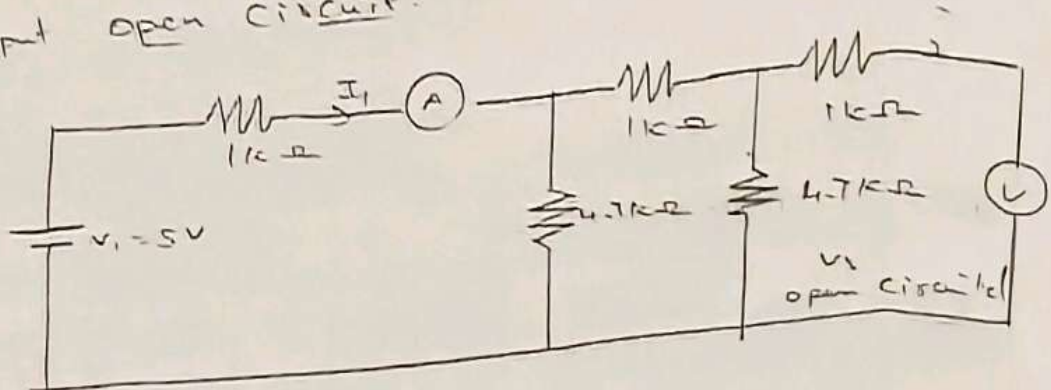
APPARATUS REQUIRED:

S-No	COMPONENTS	SPECIFICATION	QUANTITY
1.	Power Supply	-	1
2.	Multimeter	-	1
3.	Resistor	10k, 10k, 4.7k	
4.	Bread Board	-	1
5.	connecting wires	-	1
6.	PC with <u>veshti</u> multisim simulator	-	-

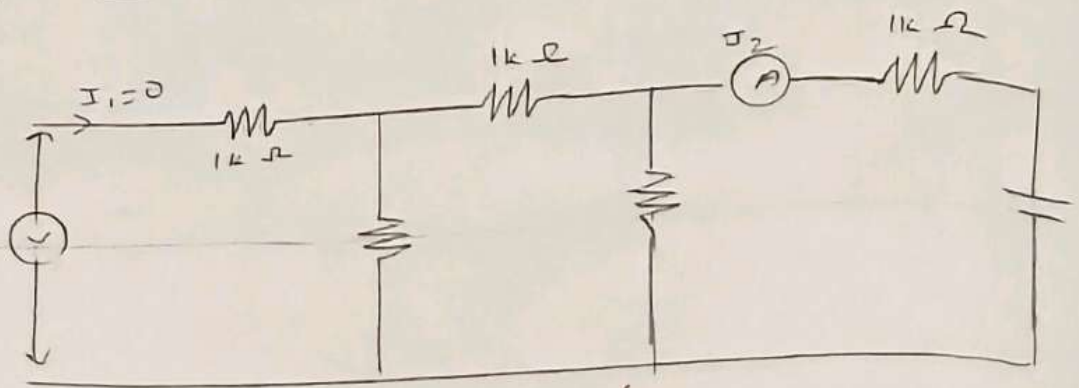
Circuit Diagram:-



Output open circuit:-



Input open circuit



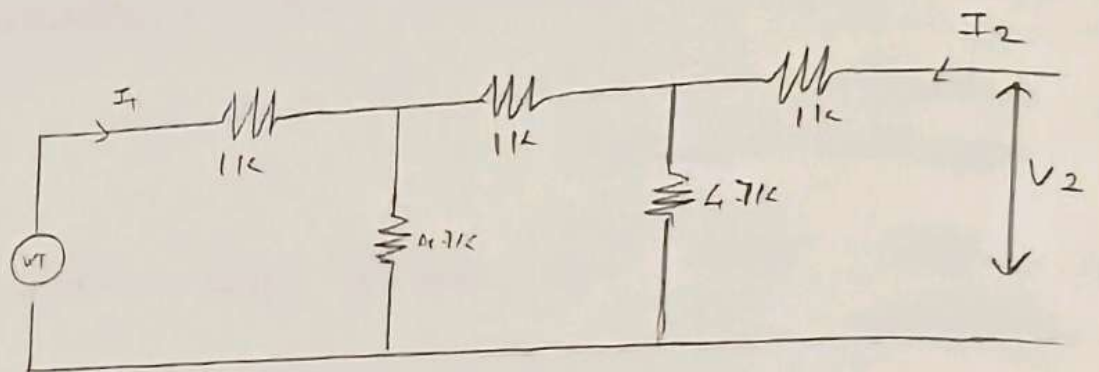
Observation

S.No	Output open Circuited $I_2 = 0$			$Z_{21} = \frac{V_2}{I_1}$ Ω	$Z_{11} = \frac{V_1}{I_1}$
	V_1	V_2	I_1		
1)	5	2	2	$Z_{21} = \frac{2}{2\text{mA}} = 1\text{k}\Omega$	$Z_{11} = \frac{5}{2\text{mA}} = 2.5\text{k}\Omega$
2)	6	3	3	$Z_{21} = \frac{3}{3\text{mA}} = 1\text{k}\Omega$	$Z_{11} = \frac{6}{3} = 2\text{k}\Omega$

S.No	Output Open Circuited $I_1 = 0$			$Z_{22} = \frac{V_2}{I_2}$ Ω	$Z_{12} = \frac{V_1}{I_2}$
	V_1	V_2	I_2		
1)	5	2	2	$Z_{22} = \frac{5}{2} = 2.5\text{k}\Omega$	$Z_{12} = \frac{6}{3} = 2\text{k}\Omega$
2)	6	3	3	$Z_{22} = \frac{3}{3} = 1\text{k}\Omega$	$Z_{12} = \frac{5}{2} = 2.5\text{k}\Omega$

PROCEDURE:-

1. Connect the circuit as shown in Fig and switch 'ON' the experiment.
2. First open the O/P terminal and supply 5V to I/P terminal. Measure O/P voltage and I/P current.
3. Secondly, open I/P terminal and supply 5V to O/P terminal. Measure I/P voltage and O/P terminal current using multimeter.
4. Calculate the values of Z Parameter using equation 1 and 2.
5. Switch 'OFF' the supply after taking the readings.



Theoretical Calculation -

$$Z_{T1} = \frac{V_1}{I_1} \Big|_{I_2=0}$$

$$= \frac{5}{2\text{mA}}$$

$$= 2.5\text{k}\Omega$$

$$Z_{T2} = \frac{V_2}{I_2} \Big|_{I_2=0}$$

$$= \frac{2}{2\text{mA}} = 1\text{k}\Omega$$

$$Z_{12} = \frac{V_1}{I_2} \Big|_{I_1=0} = \frac{2}{2\text{mA}} = 1\text{k}\Omega$$

$$Z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0} = \frac{5}{2\text{mA}} = 2.5\text{k}\Omega$$

$$Z_o = \begin{bmatrix} 2.5\text{k}\Omega & 1\text{k}\Omega \\ 1\text{k}\Omega & 2.5\text{k}\Omega \end{bmatrix}$$

S.No	DESCRIPTION	MARKS ALLOTTED	MARKS OBTAINED
1.	Experimental setup connections	20	20
2.	Observations	20	20
3.	Calculations	20	20
4.	Viva	20	18
5.	Record	20	18
			96

RESULT :

The 2-Parameter of the two part network has been calculated and verified by using Multisim simulator.